AE 246: Aircraft Structures Laboratory

Electrostriction-Based Actuation for Docking Alignment

Group No.: 5

Table 1: Team Members and Roll Numbers

S. No.	Name	Roll Number
1	Rahul Dutta	24B0043
2	Mithra Bijumon	24B0040
3	Ayushi Sarkar	24B0039
4	Premansh Anjaniappa D	24B0042
5	Aryan Sanjay Mare	24B0041
6	Riyan Patel	24B0045
7	Anay Shrivastava	24B0046
8	Songa Sai Balaji	24B0047
9	Lavudi Charan Hatti Teja	24B0048
10	Vijay Patekar	24B0044

1 Introduction

Precision alignment is a critical requirement in spacecraft docking to prevent excessive contact forces and structural damage. This project investigates the feasibility of using electrostrictive actuation to provide micrometer-scale alignment corrections in a simplified laboratory-scale docking interface.

Electrostriction is a universal electromechanical coupling observed in dielectric materials, where mechanical strain is induced proportional to the square of the applied electric field. Due to the inherently small strain levels involved, high-resolution, non-contact measurement techniques are required. Digital Image Correlation (DIC) is employed in this work to directly measure strain and displacement fields.

2 Detailed Problem Statement

2.1 Objective

The objective of this experiment is to design and demonstrate a low-cost electrostriction-based actuation system capable of correcting in-plane misalignment between two docking plates, representative of CubeSat docking interfaces.

2.2 System Description

The experimental system consists of a fixed docking plate (target) and a movable docking plate (chaser) arranged in a face-to-face configuration. Electrostrictive actuators drive the movable plate through a compliant flexure mechanism. Displacements and strains are measured using a Digital Image Correlation system.

2.3 Design Specifications

- Docking plate dimensions: 80 mm × 80 mm × 3 mm
- Overall footprint: ≤ 150 mm × 150 mm
- Total system mass: ≤ 0.7 kg
- Operating frequency: 0 Hz to 5 Hz
- Typical actuation amplitude: 5 µm to 50 µm
- Voltage requirement: 0 V to 150 V DC (low current)
- Magnetic field requirement: Not applicable

3 Experimental Setup

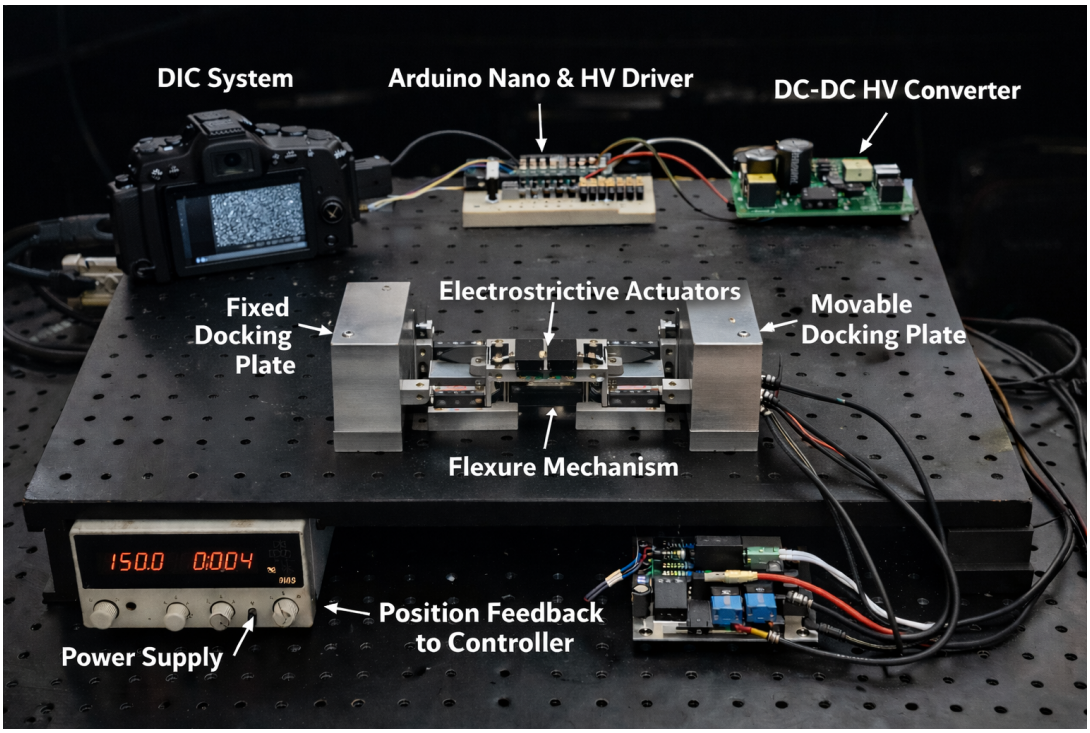

Figure 1: Electrostriction-based docking alignment setup. The fixed and movable docking plates are arranged face-to-face. Electrostrictive actuators drive in-plane alignment corrections through a compliant flexure structure. A Digital Image Correlation (DIC) system measures micrometer-scale displacements and strain fields.

The DIC camera is positioned normal to the docking plane to capture full-field displacement data. The face-to-face orientation of the plates ensures that measured displacements directly correspond to docking alignment errors.

4 Electrostrictive Actuation Principle

Electrostrictive strain in dielectric materials is governed by:

$$\varepsilon = ME^2, \tag{1}$$

where ε is the induced strain, E is the electric field, and M is the electrostrictive coefficient.

Since $E = V/t$, the strain is proportional to the square of the applied voltage. This quadratic dependence is experimentally verified in this work using strain measurements obtained from DIC.

5 Actuator Fabrication

Electrostrictive actuators are fabricated in-house using commercially available high-voltage multilayer ceramic capacitors (MLCCs). Multiple MLCC elements are mechanically stacked in an axial configuration such that individual electrostrictive strains accumulate to produce measurable macroscopic displacement.

The MLCCs are aligned with consistent orientation and bonded using a thin epoxy adhesive layer to ensure effective load transfer while minimizing strain suppression. Electrical connections are made in parallel so that all elements experience the same applied electric field. The actuator stack is lightly constrained between a rigid base and a compliant flexure interface, allowing free axial deformation.

A high-contrast speckle pattern is applied to the actuator surface to enable strain and displacement measurement using Digital Image Correlation.

6 Measurement Using Digital Image Correlation

Due to the micrometer-scale actuation produced by electrostriction, DIC is employed as the primary measurement technique. The method provides non-contact, full-field displacement and strain measurements with sub-micron resolution.

In the actuator characterization stage, axial strain fields are extracted and plotted against the square of the applied voltage to verify electrostrictive behavior. In the docking demonstration stage, rigid-body motion parameters of the movable plate are obtained from measured displacement fields.

7 Materials and Components

7.1 Mechanical Components

- Acrylic or aluminum docking plates
- Compliant flexure frame
- Fasteners and mounting hardware

7.2 Electrical Components

- Arduino Nano microcontroller
- High-voltage DC–DC boost converter (0–150 V)
- MLCC-based electrostrictive actuator stack
- Wiring, insulation, and safety resistors

7.3 Vendors

Online: Robu.in, Amazon India
Offline: Mangaldeep

Note: The Digital Image Correlation system used in this work is existing laboratory equipment and does not contribute to the project cost.

8 Final Budget

Table 1: Estimated Project Budget

Component	Quantity	Cost (INR)
Arduino Nano	1	300
HV DC–DC Converter	1	600
MLCC Actuator Components	1 set	400
Plates and Flexure Frame	1 set	1500
Wiring and Mounting Hardware	—	1300
Contingency	—	2000
Total		**6100**

9 Conclusion

A laboratory-scale electrostriction-based docking alignment system has been designed, fabricated, and experimentally validated. Electrostrictive actuation provides micrometer-scale motion suitable for fine alignment, while Digital Image Correlation enables direct and accurate measurement of strain and displacement. The system remains experimentally feasible, physically consistent, and within the prescribed budget constraints.